Open Science Literacy in the Context of the Digital Humanities

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UNESCO defined Open Science in the Recommendation published in 2021 as an "inclusive construct that combines various movements and practices aiming to make multilingual scientific knowledge openly available, accessible and reusable for everyone, to increase scientific collaborations and sharing of information for the benefits of science and society" (UNESCO, 2021).

Some authors (Bartling & Friesike, 2014; Leonelli et al., 2015; Arys et al. 2018, Schöpfel et al., 2019) suggested that Open Science can only become a reality with a cultural change. That change can be achieved by raising researchers' awareness of the importance of openness and the consequent development of policies that value and recognise open practices and training for the development of competences and skills for Open Science.

The development of competences and skills for Open Science can be referred to as Open Science Literacy, which we define as "a set of knowledge, skills and attitudes that enable researchers to: identify and adopt open research practices, develop studies based on workflows characterised by the adoption of open methods and tools, use open platforms to search for and share research results and data, and conduct experiments, observations and a range of research procedures based on open structures and platforms" (Santos, 2024).

This literacy is based on a set of interrelated elements that combine the competences and skills for Open Science with an understanding of what science is and how it is done (scientific literacy) and the ability to relate autonomously to new communication tools and to search for and use information in a multi-connected environment (digital literacy and information literacy). Those literacies are closely linked to data literacy, which is even more necessary in the context of Open Science, especially due to the importance given to the management, reuse and open sharing of research data. For this reason, it is impossible to talk about an Open Science Literacy without data literacy, or more specifically, what Schneider (2013) calls research data literacy. Finally, the mastery of research topics and trends, methods and workflows in the researcher's field of knowledge (research field mastery) is essential to develop the ability to adapt them to the principles of Open Science. Figure 1 summarizes the elements of Open Science Literacy.

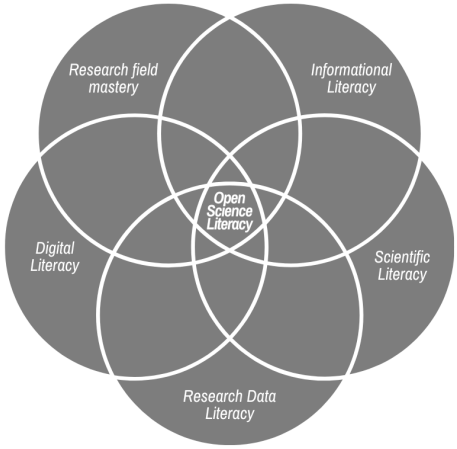

Figure 1- Open Science Literacy

In this scenario, a previous research has already been carried out in the context of a PhD thesis. That research aimed to identify initiatives to develop an Open Science Literacy in Information Science Phd candidates in four Doctoral Programs in Brazil and Portugal, and concluded that although there is Open Science training happening in these programmes, and this training is reflecting positively in the familiarity of that students have regarding Open Science and correlated topics, these initiatives are still very limited to theory and not so much to research practices aligned with Open Science principles. In order to keep the track of this previous research, this new path intends to identify, through a literature review, a set of elements that could fit the presented Open Science Literacy scheme and propose a Digital Humanities Open Science Literacy Chart of Competencies.

This Open Science Literacy, as defined here, is seen as a fundamental literacy for researchers in the context of the Digital Humanities, specially

for the Early Research Careers. Therón & Wandl-Vogt (2016) point out that "the actual set of skills that students and practitioners of Digital Humanities must be expanded and updated to consider the needs of the present socio-cultural context". In the same text, these authors state that: "Digital humanities is embracing open science and citizen science through an increasing number of projects. However, the accumulated experience is still quite shallow compared to other scientific fields".

To Smedt et al. (2018), "DH transcends discipline boundaries through a methodological commons where data and tools are, to the largest possible extent, accessible for all to use in a spirit of Open Science". Knöchelmann (2019) argues "that there needs to be a dedicated discourse for more open research and communication practices in the humanities" which the author calls open humanities.

Scholars in the humanities need to formulate what unifies them and provide points of reference about what is best for them from the perspective of their scholarship. Open humanities could be a starting point to achieve this. (Knöchelmann, 2020).

In the specific case of this study, the dimension of the "Research Field mastery" relates directly with Digital Humanities and represents the need for the researchers to be fully aware of research methods and flows, even the less usual ones, which makes the researcher more likely to develop holistic thinking when designing their research. That can contribute to the development and adoption of research methods and flows better aligned with the principles of Open Science.

Scientific literacy contributes to the Open Science scenario in the sense that a researcher with a high level of scientific literacy is certainly better able to identify, select and even create more reproducible, transparent and sustainable research methods, which in turn are also more open. Particularly in the digital humanities, scientific literacy includes not only the ability to identify what science is, but also to understand the scientific method (and how to use it to solve problems), and to distinguish between scientific and non-scientific content, but also deeper notions such as critical thinking about science and ways of engaging with scientific knowledge. It is therefore a fundamental skill in the training of any scientist, especially when they are exposed to a scenario of intense change in the way science is done and communicated, as is the case today. This also includes issues related to ethics and integrity in science, which are being discussed very pertinently around the OS.

In addition, the increase of the scientific and/or academic competences gets even more relevant, especially considering issues such as the communication of science and the mastery of research methods, making academic literacy an important set of competences in the Open Science scenario. One of the impacts felt in relation to scientific communication is the addition of new elements throughout the research life cycle, from its design to the evaluation metrics adopted to recognise research results. This is why it is necessary to include these notions in the preparation of young researchers, such as pre-registration of hypotheses, data sharing and dissemination of preprints. It is understood that Open Science changes the way many processes are communicated and recorded, formalising various stages that were previously not shared, such as the recording of research projects and the raw data resulting from the work. This new flow, therefore, does not change the essence of research; its main difference from the previous process is transparency, reuse, collaboration and social inclusion in scientific production. (Silva & Silveira, 2019, p.3)

We cannot conceive an Open Science Literacy without considering the main phenomenon that strengthened the discussions about Open Science: the digitisation of science, in all scientific fields, even more so in the Digital Humanities. For this reason, the need for a higher level of digital literacy goes beyond simply searching for information on the web, but also includes different forms of reading, far beyond the textual, and much more accurate creative skills.

As previously mentioned, this research, as a new path born from a PhD thesis, intends to identify a set of elements to fit the proposed Open Science Literacy scheme focused in the needs within Digital Humanities. We believe the establishment of a set of Open Science Competencies specific for the Digital Humanities will be one of these efforts to enrich Open Humanities and contribute especially to the development of Open Science competencies for the Early Careers Research in the context of doctoral programs curriculum.

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